

Qualification Pack



Optoelectronic Technology

QP Code: ELE/Q6706

Version: 1.0

NSQF Level: 4.5

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ELE/Q6706: Optoelectronic Technology

Brief Job Description

This comprehensive training program equips participants with the skills to design, install, test, maintain, and optimize optoelectronic systems, including LEDs, sensors, displays, and photovoltaic applications. Covering Health & Safety, customer interaction, system design, installation, maintenance, and energy optimization, the program combines theoretical knowledge with hands-on practical experience using industry-standard tools and software. Participants will gain the expertise to deliver efficient, safe, and sustainable optoelectronic solutions in professional settings.

Personal Attributes

Detail-oriented, safety-conscious, and technically proficient professional with strong problem-solving abilities, hands-on practical skills, and effective communication for delivering efficient and sustainable optoelectronic solutions.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [ELE/N6725: Work organization and management \(Optoelectronic Technology\)](#)
2. [ELE/N6726: Communication and interpersonal skills \(Optoelectronic Technology\)](#)
3. [ELE/N6727: Design and manufacturing of optoelectronic applications \(Optoelectronic Technology\)](#)
4. [ELE/N6728: Installation and implementation of optoelectronic applications \(Optoelectronic Technology\)](#)
5. [ELE/N6729: Maintenance of optoelectronic applications of security systems \(Optoelectronic Technology\)](#)
6. [ELE/N6730: Optimization of optoelectronic applications \(Optoelectronic Technology\)](#)

Qualification Pack (QP) Parameters

Sector	Electronics
Sub-Sector	Communication and Broadcasting
Occupation	Product Design & Development-EM&B
Country	India

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NSQF Level	4.5
Credits	17
Aligned to NCO/ISCO/ISIC Code	NCO-2015/7422.1203
Minimum Educational Qualification & Experience	No formal education prescribed
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NA
Minimum Job Entry Age	16 Years
Last Reviewed On	NA
Next Review Date	06/02/2028
NSQC Approval Date	06/02/2026
Version	1.0
Reference code on NQR	QG-4.5-EH-05196-2026-V1-ESSCI
NQR Version	1

Remarks:

NA

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ELE/N6725: Work organization and management (Optoelectronic Technology)

Description

This NOS is about to plan, organize, and prioritize tasks effectively to ensure timely design, installation, and maintenance of optoelectronic systems while.

Scope

The scope covers the following :

- Work organization and management

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Apply and comply with health, safety, and environmental (HSE) regulations, legislation, and documentation in all workshop operations.
- PC2.** Follow electrical safety procedures diligently while handling electronics and electrical systems.
- PC3.** Identify, select, and correctly use personal protective equipment (PPE) including safety footwear, eye, and ear protection.
- PC4.** Select, use, maintain, and store all tools and equipment safely and according to standard operating procedures.
- PC5.** Select, handle, and store all materials safely, including expensive or vulnerable fixtures/fittings, ensuring minimal damage.
- PC6.** Plan and organize the work area to maximize efficiency, maintain cleanliness, and uphold discipline in workspace tidiness.
- PC7.** Apply accurate measurement techniques and adhere to workflow principles to ensure precision and quality.
- PC8.** Manage work schedules effectively to meet deadlines and optimize productivity while maintaining high quality.
- PC9.** Monitor work progress regularly, check outcomes, and maintain quality standards consistently.
- PC10.** Implement sustainable practices, including the use of green materials, recycling, and minimizing wastage to manage costs.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Knowledge of health, safety, and environmental (HSE) regulations, relevant legislation, documentation requirements, and compliance procedures applicable to workshop and electrical operations.

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- KU2.** Understanding of electrical safety principles, hazard identification methods, risk assessment procedures, and safe handling practices for electronic and electrical systems.
- KU3.** Knowledge of proper selection, use, maintenance, and storage of personal protective equipment (PPE), tools, equipment, and materials to prevent accidents and damage.
- KU4.** Understanding of workplace organization principles, workflow management, measurement techniques, quality control standards, and productivity optimization methods.
- KU5.** Knowledge of sustainable practices including green material usage, waste minimization strategies, recycling procedures, and cost-effective resource management.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Apply and consistently comply with HSE regulations and electrical safety procedures to ensure a safe and legally compliant working environment.
- GS2.** Select and use appropriate PPE, tools, equipment, and materials safely while maintaining them according to standard operating procedures.
- GS3.** Organize and maintain a clean, efficient, and disciplined workspace to enhance productivity and operational effectiveness.
- GS4.** Apply accurate measurement techniques, monitor work progress, and maintain quality standards to ensure precision and high-quality outcomes.
- GS5.** Manage time, resources, and work schedules effectively while implementing environmentally sustainable and cost-efficient practices.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	10	-	-
PC1. Apply and comply with health, safety, and environmental (HSE) regulations, legislation, and documentation in all workshop operations.	-	1	-	-
PC2. Follow electrical safety procedures diligently while handling electronics and electrical systems.	-	1	-	-
PC3. Identify, select, and correctly use personal protective equipment (PPE) including safety footwear, eye, and ear protection.	-	1	-	-
PC4. Select, use, maintain, and store all tools and equipment safely and according to standard operating procedures.	-	1	-	-
PC5. Select, handle, and store all materials safely, including expensive or vulnerable fixtures/fittings, ensuring minimal damage.	-	1	-	-
PC6. Plan and organize the work area to maximize efficiency, maintain cleanliness, and uphold discipline in workspace tidiness.	-	1	-	-
PC7. Apply accurate measurement techniques and adhere to workflow principles to ensure precision and quality.	-	1	-	-
PC8. Manage work schedules effectively to meet deadlines and optimize productivity while maintaining high quality.	-	1	-	-
PC9. Monitor work progress regularly, check outcomes, and maintain quality standards consistently.	-	1	-	-
PC10. Implement sustainable practices, including the use of green materials, recycling, and minimizing wastage to manage costs.	-	1	-	-
NOS Total	-	10	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	ELE/N6725
NOS Name	Work organization and management (Optoelectronic Technology)
Sector	Electronics
Sub-Sector	
Occupation	Product Design & Development-EM&B
NSQF Level	4.5
Credits	2
Version	1.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2028
NSQC Clearance Date	06/02/2026

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ELE/N6726: Communication and interpersonal skills (Optoelectronic Technology)

Description

This NOS is about to effectively communicate technical and non-technical information with clients and team members to ensure clarity in optoelectronic projects. Maintain professionalism, teamwork, and positive interpersonal relationships for smooth coordination and successful task completion.

Scope

The scope covers the following :

- Communication and interpersonal skills

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Interpret customer requirements accurately and manage customer expectations effectively.
- PC2.** Provide advice and guidance on products, solutions, and technological advancements to meet customer needs.
- PC3.** Create training materials and implement training according to the construction and specifications of each system.
- PC4.** Develop training content and related materials specifically for optoelectronic applications.
- PC5.** Visualize and translate customer wishes into actionable recommendations that meet or improve design and budgetary requirements.
- PC6.** Question customers deeply to fully understand their requirements and provide clear, actionable instructions.
- PC7.** Introduce and coordinate with related trades to support and fulfill customer requirements.
- PC8.** Produce written reports, including technical documentation, cost, and time estimates for customers and organizations.
- PC9.** Recognize and adapt to the changing needs of related trades to ensure smooth workflow and customer satisfaction.
- PC10.** Work effectively as a member of teams, applying principles of teamwork and conflict resolution to maintain productive working relationships.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Knowledge of customer requirement analysis techniques, expectation management strategies, and effective communication principles in technical service environments.
- KU2.** Understanding of optoelectronic products, systems, technological advancements, and solution-based advisory practices relevant to customer applications.

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- KU3.** Knowledge of training design methodologies, instructional techniques, and development of technical training materials tailored to system construction and specifications.
- KU4.** Understanding of technical documentation standards, report writing formats, cost estimation methods, and project time planning procedures.
- KU5.** Knowledge of teamwork principles, inter-trade coordination processes, workflow integration, and conflict resolution strategies in multidisciplinary project environments.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Interpret customer requirements accurately through structured questioning and translate them into practical, cost-effective, and technically sound solutions.
- GS2.** Provide professional advice and guidance on products, technological advancements, and application solutions aligned with customer needs and budget constraints.
- GS3.** Develop and deliver structured training programs and technical materials for optoelectronic systems based on system specifications and user requirements.
- GS4.** Prepare clear written reports, technical documentation, cost estimates, and project timelines to support informed decision-making by customers and stakeholders.
- GS5.** Collaborate effectively with customers, teams, and related trades while adapting to changing requirements and applying teamwork and conflict management skills.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	5	-	-
PC1. Interpret customer requirements accurately and manage customer expectations effectively.	-	-	-	-
PC2. Provide advice and guidance on products, solutions, and technological advancements to meet customer needs.	-	1	-	-
PC3. Create training materials and implement training according to the construction and specifications of each system.	-	-	-	-
PC4. Develop training content and related materials specifically for optoelectronic applications.	-	1	-	-
PC5. Visualize and translate customer wishes into actionable recommendations that meet or improve design and budgetary requirements.	-	-	-	-
PC6. Question customers deeply to fully understand their requirements and provide clear, actionable instructions.	-	1	-	-
PC7. Introduce and coordinate with related trades to support and fulfill customer requirements.	-	-	-	-
PC8. Produce written reports, including technical documentation, cost, and time estimates for customers and organizations.	-	1	-	-
PC9. Recognize and adapt to the changing needs of related trades to ensure smooth workflow and customer satisfaction.	-	-	-	-
PC10. Work effectively as a member of teams, applying principles of teamwork and conflict resolution to maintain productive working relationships.	-	1	-	-
NOS Total	-	5	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	ELE/N6726
NOS Name	Communication and interpersonal skills (Optoelectronic Technology)
Sector	Electronics
Sub-Sector	
Occupation	Product Design & Development-EM&B
NSQF Level	4.5
Credits	1
Version	1.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2028
NSQC Clearance Date	06/02/2026

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ELE/N6727: Design and manufacturing of optoelectronic applications (Optoelectronic Technology)

Description

This NOS is about to design optoelectronic systems and applications by applying engineering principles, component selection, circuit design, and performance analysis. Manufacture, assemble, and test optoelectronic devices such as LEDs, sensors, displays, and photovoltaic systems to ensure quality, efficiency, and reliability.

Scope

The scope covers the following :

- Design and manufacturing of optoelectronic applications

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Analyse the important parameters and functional requirements of the optoelectronic application environment.
- PC2.** Design optoelectronic application schemes according to industry requirements and application needs.
- PC3.** Manufacture simple optoelectronic applications following the designed schemes.
- PC4.** Inspect and ensure the manufacturing quality and operational standards of optoelectronic applications.
- PC5.** Test and select correct specifications for various optoelectronic products, including photovoltaic products, LED luminaires, optoelectronic transmission and display equipment, sensors, and laser systems.
- PC6.** Select appropriate control equipment, photovoltaic power generation systems, sensors, laser systems, application terminals, and control platforms to achieve energy-saving and efficiency goals in integrated applications.
- PC7.** Analyse the application environment to optimize manufacturing schemes, energy-saving measures, and light efficiency design schemes.
- PC8.** Select, configure, and optimize optical and electronic components, including drive and heat dissipation modules, for high-performance optoelectronic applications.
- PC9.** Test the function and performance parameters of optical and electronic components or optoelectronic products to ensure compliance with specifications.
- PC10.** Apply industry-aligned software tools (e.g., DIALux, LED simulation platforms, Arduino-based controllers) to simulate, programme, and validate lighting and sensor-based applications.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

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- KU1.** Knowledge of key parameters, environmental conditions, and functional requirements influencing optoelectronic applications, including illumination levels, energy efficiency, thermal management, and system integration factors.
- KU2.** Understanding of industry standards, design methodologies, and application-specific requirements for developing optoelectronic schemes and integrated solutions.
- KU3.** Knowledge of manufacturing processes, quality control procedures, and operational standards applicable to optoelectronic products such as photovoltaic systems, LED luminaires, sensors, laser systems, and display equipment.
- KU4.** Understanding of component selection principles, including optical elements, electronic drivers, heat dissipation modules, control platforms, and energy management systems for high-performance applications.
- KU5.** Knowledge of testing methodologies, performance parameter evaluation techniques, and industry-aligned software tools (e.g., simulation, programming, and validation platforms) used in optoelectronic system design and optimization.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Analyze application environments and functional requirements to design efficient, reliable, and industry-compliant optoelectronic solutions.
- GS2.** Develop and manufacture simple optoelectronic systems according to approved design schemes while ensuring quality and operational standards.
- GS3.** Select, configure, and optimize optical and electronic components to enhance performance, energy efficiency, and system reliability.
- GS4.** Test and evaluate functional and performance parameters of optoelectronic products to ensure compliance with specifications and industry requirements.
- GS5.** Utilize industry-relevant software tools to simulate, program, validate, and optimize lighting, photovoltaic, sensor-based, and integrated optoelectronic applications.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	20	-	-
PC1. Analyse the important parameters and functional requirements of the optoelectronic application environment.	-	2	-	-
PC2. Design optoelectronic application schemes according to industry requirements and application needs.	-	2	-	-
PC3. Manufacture simple optoelectronic applications following the designed schemes.	-	2	-	-
PC4. Inspect and ensure the manufacturing quality and operational standards of optoelectronic applications.	-	2	-	-
PC5. Test and select correct specifications for various optoelectronic products, including photovoltaic products, LED luminaires, optoelectronic transmission and display equipment, sensors, and laser systems.	-	2	-	-
PC6. Select appropriate control equipment, photovoltaic power generation systems, sensors, laser systems, application terminals, and control platforms to achieve energy-saving and efficiency goals in integrated applications.	-	2	-	-
PC7. Analyse the application environment to optimize manufacturing schemes, energy-saving measures, and light efficiency design schemes.	-	2	-	-
PC8. Select, configure, and optimize optical and electronic components, including drive and heat dissipation modules, for high-performance optoelectronic applications.	-	2	-	-
PC9. Test the function and performance parameters of optical and electronic components or optoelectronic products to ensure compliance with specifications.	-	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. Apply industry-aligned software tools (e.g., DIALux, LED simulation platforms, Arduino-based controllers) to simulate, programme, and validate lighting and sensor-based applications.	-	2	-	-
NOS Total	-	20	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	ELE/N6727
NOS Name	Design and manufacturing of optoelectronic applications (Optoelectronic Technology)
Sector	Electronics
Sub-Sector	
Occupation	Product Design & Development-EM&B
NSQF Level	4.5
Credits	4
Version	1.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2028
NSQC Clearance Date	06/02/2026

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ELE/N6728: Installation and implementation of optoelectronic applications (Optoelectronic Technology)

Description

This NOS defines the competencies required to install, integrate, test, and commission optoelectronic systems in accordance with technical specifications, safety standards, and application requirements

Scope

The scope covers the following :

- Installation and implementation of optoelectronic applications

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Read and interpret engineering drawings, wiring diagrams, schematics, technical manuals, and engineering instructions accurately.
- PC2.** Analyze customer requirements for optoelectronic system applications and identify functional and efficiency requirements.
- PC3.** Select appropriate optical and electronic components according to application needs, functional parameters, and efficiency design schemes.
- PC4.** Test relevant parameters and characteristics of optical and electronic components, sensors, laser systems, and control devices.
- PC5.** Select and configure suitable optoelectronic application schemes according to demand analysis and application environment.
- PC6.** Select, configure, and optimize drive and heat dissipation modules based on application terminal structure and power requirements.
- PC7.** Provide system structure drawings, construction drawings, and related technical documentation for installations.
- PC8.** Install and commission equipment, components, devices, upgrades, or refurbished optoelectronic equipment.
- PC9.** Use a variety of optoelectronic display devices, LED panels, screens, sensors, and photovoltaic power generation equipment effectively.
- PC10.** Select energy-saving control methods and test the characteristics of optoelectronic application equipment and system control platforms
- PC11.** Install and perform basic configuration of opto sensors and cameras, including connection, alignment, and computer-based setup.
- PC12.** Install, debug, and operate the optoelectronic application system, ensuring normal operation and functionality.
- PC13.** Programme or configure Arduino-based smart lighting control systems and basic image display devices.
- PC14.** Programme or configure Arduino-based smart lighting control systems and basic image display devices.

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- PC15.** Use electronic common tools and optoelectronic-specific instruments, such as soldering irons, digital multimeters, and chroma meters, to carry out installation, testing, and configuration.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Comprehensive knowledge of engineering drawings, wiring diagrams, circuit schematics, system layouts, and technical manuals to accurately interpret installation, configuration, and operational requirements.
- KU2.** In-depth understanding of optical and electronic components such as LEDs, laser systems, sensors, cameras, photovoltaic modules, control devices, and their functional parameters, efficiency ratings, and application suitability.
- KU3.** Knowledge of analyzing customer requirements, environmental conditions, power demands, thermal management needs, and structural constraints for selecting and configuring appropriate optoelectronic system schemes.
- KU4.** Understanding of testing methodologies, performance characteristics, calibration techniques, and parameter measurement of optical and electronic components using appropriate instruments and diagnostic tools.
- KU5.** Knowledge of installation procedures, commissioning processes, energy-saving control methods, Arduino-based smart lighting systems, system debugging techniques, and preparation of technical documentation and test reports.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Ability to read, interpret, and implement engineering drawings, schematics, manuals, and technical instructions accurately while analyzing system requirements and identifying functional and efficiency needs.
- GS2.** Proficiency in installing, commissioning, configuring, debugging, and operating optoelectronic systems, including sensors, cameras, display devices, LED panels, and photovoltaic systems.
- GS3.** Competence in using electronic tools and optoelectronic instruments such as soldering irons, digital multimeters, chroma meters, and testing equipment for installation, calibration, and system validation.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	35	-	-
PC1. Read and interpret engineering drawings, wiring diagrams, schematics, technical manuals, and engineering instructions accurately.	-	2	-	-
PC2. Analyze customer requirements for optoelectronic system applications and identify functional and efficiency requirements.	-	2	-	-
PC3. Select appropriate optical and electronic components according to application needs, functional parameters, and efficiency design schemes.	-	2	-	-
PC4. Test relevant parameters and characteristics of optical and electronic components, sensors, laser systems, and control devices.	-	2	-	-
PC5. Select and configure suitable optoelectronic application schemes according to demand analysis and application environment.	-	2	-	-
PC6. Select, configure, and optimize drive and heat dissipation modules based on application terminal structure and power requirements.	-	2	-	-
PC7. Provide system structure drawings, construction drawings, and related technical documentation for installations.	-	2	-	-
PC8. Install and commission equipment, components, devices, upgrades, or refurbished optoelectronic equipment.	-	2	-	-
PC9. Use a variety of optoelectronic display devices, LED panels, screens, sensors, and photovoltaic power generation equipment effectively.	-	2	-	-
PC10. Select energy-saving control methods and test the characteristics of optoelectronic application equipment and system control platforms	-	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. Install and perform basic configuration of opto sensors and cameras, including connection, alignment, and computer-based setup.	-	3	-	-
PC12. Install, debug, and operate the optoelectronic application system, ensuring normal operation and functionality.	-	3	-	-
PC13. Programme or configure Arduino-based smart lighting control systems and basic image display devices.	-	3	-	-
PC14. Programme or configure Arduino-based smart lighting control systems and basic image display devices.	-	3	-	-
PC15. Use electronic common tools and optoelectronic-specific instruments, such as soldering irons, digital multimeters, and chroma meters, to carry out installation, testing, and configuration.	-	3	-	-
NOS Total	-	35	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	ELE/N6728
NOS Name	Installation and implementation of optoelectronic applications (Optoelectronic Technology)
Sector	Electronics
Sub-Sector	
Occupation	Product Design & Development-EM&B
NSQF Level	4.5
Credits	6
Version	1.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2028
NSQC Clearance Date	06/02/2026

Qualification Pack

ELE/N6729: Maintenance of optoelectronic applications of security systems (Optoelectronic Technology)

Description

This NOS is about to define the competencies required to inspect, troubleshoot, service, and maintain optoelectronic-based security systems to ensure reliable performance and compliance with safety standards.

Scope

The scope covers the following :

- Maintenance of optoelectronic applications of security systems

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Read and interpret construction drawings, technical data, and documents for optoelectronic application systems.
- PC2.** Analyze the working principles of optoelectronic applications, signal transmission processes, and characteristic parameters of each system component.
- PC3.** Identify the causes of faults in optoelectronic systems and select appropriate fault detection schemes based on fault phenomena.
- PC4.** Diagnose and replace faulty equipment, modules, lines, parts, and components in optoelectronic applications.
- PC5.** Use standard optoelectronic testing tools, instruments, and maintenance equipment safely and effectively.
- PC6.** Perform maintenance on each module of optoelectronic application systems to ensure proper operation.
- PC7.** Modify control platform parameters to optimize system performance after maintenance operations.
- PC8.** Debug and optimize optoelectronic application systems to ensure correct functionality.
- PC9.** Test optoelectronic applications after maintenance to verify proper operation and performance.
- PC10.** Apply diagnostic methods to identify and correct errors in sensor/camera configurations and smart lighting control systems.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Comprehensive knowledge of construction drawings, system layouts, wiring diagrams, and technical documentation to accurately interpret optoelectronic application system requirements.

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- KU2.** In-depth understanding of optoelectronic application principles, including light generation and detection, signal transmission processes, control logic, and characteristic parameters of system components such as sensors, cameras, LED drivers, and control modules.
- KU3.** Knowledge of common faults in optoelectronic systems, root cause analysis techniques, troubleshooting procedures, and selection of appropriate fault detection schemes based on observed fault phenomena.
- KU4.** Understanding of standard optoelectronic testing tools and instruments such as digital multimeters, oscilloscopes, signal testers, optical power meters, and maintenance equipment, including their safe operation and calibration requirements.
- KU5.** Knowledge of maintenance strategies, parameter adjustment of control platforms, debugging techniques, and performance optimization methods for sensors, cameras, and smart lighting control systems.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Ability to read and interpret construction drawings and technical data, analyze system working principles, and understand component-level interactions within optoelectronic applications.
- GS2.** Skill in identifying fault symptoms, selecting suitable diagnostic approaches, isolating defective modules, and determining corrective actions efficiently.
- GS3.** Proficiency in safely diagnosing, repairing, replacing, and maintaining faulty equipment, modules, wiring lines, and components to restore system functionality.
- GS4.** Competence in using standard testing tools and instruments to verify system performance, modify control parameters, debug systems, and optimize functionality after maintenance.
- GS5.** Ability to follow safety procedures, use tools responsibly, document maintenance activities, prepare test reports, and ensure reliable and efficient system operation in compliance with industry standards.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	15	-	-
PC1. Read and interpret construction drawings, technical data, and documents for optoelectronic application systems.	-	2	-	-
PC2. Analyze the working principles of optoelectronic applications, signal transmission processes, and characteristic parameters of each system component.	-	2	-	-
PC3. Identify the causes of faults in optoelectronic systems and select appropriate fault detection schemes based on fault phenomena.	-	2	-	-
PC4. Diagnose and replace faulty equipment, modules, lines, parts, and components in optoelectronic applications.	-	2	-	-
PC5. Use standard optoelectronic testing tools, instruments, and maintenance equipment safely and effectively.	-	2	-	-
PC6. Perform maintenance on each module of optoelectronic application systems to ensure proper operation.	-	1	-	-
PC7. Modify control platform parameters to optimize system performance after maintenance operations.	-	1	-	-
PC8. Debug and optimize optoelectronic application systems to ensure correct functionality.	-	1	-	-
PC9. Test optoelectronic applications after maintenance to verify proper operation and performance.	-	1	-	-
PC10. Apply diagnostic methods to identify and correct errors in sensor/camera configurations and smart lighting control systems.	-	1	-	-
NOS Total	-	15	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	ELE/N6729
NOS Name	Maintenance of optoelectronic applications of security systems (Optoelectronic Technology)
Sector	Electronics
Sub-Sector	
Occupation	Product Design & Development-EM&B
NSQF Level	4.5
Credits	2
Version	1.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2028
NSQC Clearance Date	06/02/2026

Qualification Pack

ELE/N6730: Optimization of optoelectronic applications (Optoelectronic Technology)

Description

This NOS is about to defines the competencies required to analyze, calibrate, and enhance optoelectronic systems to ensure optimal performance, efficiency, and reliability in application environments.

Scope

The scope covers the following :

- Optimization of optoelectronic applications

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Analyze existing optoelectronic systems and make appropriate optimizations based on legacy or traditional designs.
- PC2.** Identify optoelectronic applications or products that are not sustainable or may be harmful to people's health and propose adjustments.
- PC3.** Select and implement optimization schemes for optoelectronic systems to improve efficiency and safety.
- PC4.** Achieve energy saving within defined operational requirements and parameters.
- PC5.** Optimize the heat dissipation performance of optoelectronic products to ensure longevity and efficiency.
- PC6.** Design and optimize the structural configuration of optoelectronic products for improved performance.
- PC7.** Optimize driving circuits of optoelectronic products to improve operational efficiency and reduce power loss.
- PC8.** Design and implement light efficiency schemes for optical and electronic components in optoelectronic products.
- PC9.** Improve the user experience of optoelectronic products by integrating optimized design features.
- PC10.** Enhance conversion efficiency of photovoltaic systems and reduce overall energy consumption of optoelectronic applications.
- PC11.** Develop energy-saving strategies for optoelectronic application systems and integrate them into operational workflows.
- PC12.** Upgrade existing traditional lighting systems to intelligent lighting systems to optimize energy use.
- PC13.** Optimize energy use and control strategies for smart lighting and sensor-based applications.
- PC14.** Use relevant software tools to simulate application scenarios and optimize optical and electronic component configurations.
- PC15.** Apply optimized design drawings to configure systems so they operate normally, efficiently, and sustainably.

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Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Comprehensive knowledge of traditional and existing optoelectronic system architectures, including structural configurations, driving circuits, thermal management, and performance limitations for effective optimization.
- KU2.** Understanding of environmental impact, energy efficiency standards, heat generation effects, optical radiation safety, and potential health hazards associated with optoelectronic applications.
- KU3.** Knowledge of energy-saving methodologies, photovoltaic conversion efficiency, intelligent lighting systems, sensor-based control strategies, and reduction of power losses in driving circuits.
- KU4.** Understanding of heat dissipation mechanisms, material selection, structural design improvements, and performance optimization techniques to enhance product longevity and operational stability.
- KU5.** Knowledge of relevant software tools for simulation, modeling, and performance analysis of optical and electronic components, along with preparation and implementation of optimized design drawings.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Ability to evaluate existing optoelectronic systems, identify inefficiencies or safety risks, and propose technically sound optimization schemes for improved efficiency and sustainability.
- GS2.** Skill in designing and implementing energy-saving strategies, upgrading conventional lighting to intelligent systems, and enhancing photovoltaic and system conversion efficiency.
- GS3.** Proficiency in optimizing driving circuits, improving heat dissipation performance, configuring structural designs, and reducing power loss in optoelectronic products.
- GS4.** Competence in using software tools to simulate real-world application scenarios, optimize optical/electronic component configurations, and implement refined system designs effectively.
- GS5.** Ability to integrate optimized design features that enhance user experience, ensure safe and sustainable operation, reduce overall energy consumption, and maintain compliance with operational requirements.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	15	-	-
PC1. Analyze existing optoelectronic systems and make appropriate optimizations based on legacy or traditional designs.	-	1	-	-
PC2. Identify optoelectronic applications or products that are not sustainable or may be harmful to people's health and propose adjustments.	-	1	-	-
PC3. Select and implement optimization schemes for optoelectronic systems to improve efficiency and safety.	-	1	-	-
PC4. Achieve energy saving within defined operational requirements and parameters.	-	1	-	-
PC5. Optimize the heat dissipation performance of optoelectronic products to ensure longevity and efficiency.	-	1	-	-
PC6. Design and optimize the structural configuration of optoelectronic products for improved performance.	-	1	-	-
PC7. Optimize driving circuits of optoelectronic products to improve operational efficiency and reduce power loss.	-	1	-	-
PC8. Design and implement light efficiency schemes for optical and electronic components in optoelectronic products.	-	1	-	-
PC9. Improve the user experience of optoelectronic products by integrating optimized design features.	-	1	-	-
PC10. Enhance conversion efficiency of photovoltaic systems and reduce overall energy consumption of optoelectronic applications.	-	1	-	-
PC11. Develop energy-saving strategies for optoelectronic application systems and integrate them into operational workflows.	-	1	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. Upgrade existing traditional lighting systems to intelligent lighting systems to optimize energy use.	-	1	-	-
PC13. Optimize energy use and control strategies for smart lighting and sensor-based applications.	-	1	-	-
PC14. Use relevant software tools to simulate application scenarios and optimize optical and electronic component configurations.	-	1	-	-
PC15. Apply optimized design drawings to configure systems so they operate normally, efficiently, and sustainably.	-	1	-	-
NOS Total	-	15	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ELE/N6730
NOS Name	Optimization of optoelectronic applications (Optoelectronic Technology)
Sector	Electronics
Sub-Sector	
Occupation	Product Design & Development-EM&B
NSQF Level	4.5
Credits	2
Version	1.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2028
NSQC Clearance Date	06/02/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training centre (as per assessment criteria below).

Qualification Pack

5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/training centre based on this criterion.

6. To pass the Qualification Pack, every trainee should score a minimum of 70% of aggregate marks to successfully clear the assessment.

7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
ELE/N6725.Work organization and management (Optoelectronic Technology)	-	10	-	-	10	10
ELE/N6726.Communication and interpersonal skills (Optoelectronic Technology)	-	5	-	-	5	5
ELE/N6727.Design and manufacturing of optoelectronic applications (Optoelectronic Technology)	-	20	-	-	20	20
ELE/N6728.Installation and implementation of optoelectronic applications (Optoelectronic Technology)	-	35	-	-	35	35
ELE/N6729.Maintenance of optoelectronic applicationsof security systems (Optoelectronic Technology)	-	15	-	-	15	15

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National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
ELE/N6730.Optimization of optoelectronic applications (Optoelectronic Technology)	-	15	-	-	15	15
Total	0	100	-	-	100	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.